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A Fuzzy Clustering and Association Rule-based Framework for Understanding Cluster-specific Vulnerabilities in Epidemic Modeling

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Abstract. The emergence of epidemics caused by contagious pathogens underscores the critical need for robust epidemiological modeling to understand the multifaceted drivers of disease spread and to inform timely intervention strategies. In this study, we propose a data-driven approach that leverages fuzzy *c*-means soft clustering to group regions with similar epidemiological characteristics based on climatic, demographic, and disease-related data. This method accommodates overlapping group memberships, reflecting the dynamic and uncertain nature of real-world outbreaks. To assess the epidemiological risk associated with each cluster, we analyze the contributing factors and explore their influence on disease severity. Furthermore, we employ fuzzy association rule mining to uncover meaningful relationships between environmental and demographic variables and epidemic outcomes. Using COVID-19 as a case study, we demonstrate the effectiveness of this framework in revealing region-specific patterns and underlying risk factors. Our findings contribute to a deeper understanding of epidemic dynamics and provide valuable insights to support evidence-based public health decision-making.

Keywords: fuzzy *c*-means clustering, epidemiological risk, fuzzy association rule mining

1 Introduction

Throughout history, infectious disease outbreaks have profoundly affected humanity, ranging from the Black Death [9] in the 14th century to more recent pandemics such as Influenza [28], Measles [4], Ebola [8] and the globally widespread COVID-19 [23]. On the one hand, advances in medical science, the development of effective vaccines, and timely intervention strategies such as lockdown and social distancing have greatly improved our ability to control pandemics. However, the emergence of antibiotic resistant pathogens [32] and the frequent mutation of infectious strains have made the management of outbreaks even more challenging [27].

In this context, epidemic modeling plays a vital role in identifying high-risk areas, uncovering factors driving infection spread, predicting future outbreaks, and guiding effective intervention strategies. Classical compartmental models such as SIR [33], SEIR [24], and SIRD [30] are widely used but often struggle with accuracy due to their reliance on simplifying assumptions [22]. Recent advances in machine learning (ML) and deep learning (DL) have significantly improved epidemic prediction. ML techniques such as decision trees, random forests, and SVMs offer flexible data-driven approaches [2] [18], while DL models like Convolutional Neural Networks (CNNs) [11] and Recurrent Neural Networks (RNNs) [35] have shown great potential in solving various epidemiological prediction challenges. In addition, graph-based deep learning models, such as Graph Neural Networks (GNNs), have emerged as powerful tools for epidemic forecasting, helping predict future infections or deaths [13] [25].

The primary objective of the current research is not to forecast the numerical outcomes of an epidemic. Instead, it focuses on identifying groups of regions similarly affected on a daily basis and analyzing how their cluster memberships evolve over time. Using fuzzy clustering, we examine key epidemic-related factors and assess the epidemiological risk of each cluster. While some prior studies have explored hidden community structures using unsupervised learning [7] [6], few have applied fuzzy clustering, and even fewer have integrated pandemic data with climatic and demographic characteristics. Soft clustering methods, increasingly favored over traditional hard clustering [12] [29], allow regions to belong to multiple clusters with varying degrees of membership—reflecting the overlapping influence of environmental and demographic factors. This flexibility makes fuzzy clustering particularly suitable for modeling the complex, dynamic nature of epidemics.

To enhance the analysis of epidemic clusters, we employ fuzzy association rule mining to identify key climatic and demographic factors driving outbreaks within each cluster. Unlike traditional methods such as the Apriori algorithm [17], which are designed for categorical data, fuzzy association rule mining is better suited for our context, where all attributes—pandemic statistics, climatic factors (e.g., temperature, humidity), and demographic characteristics (e.g., population density)—are continuous. While our previous work explored general associations between COVID-19 and environmental factors, this study goes further by extracting cluster-specific rules. These rules reveal how the drivers of outbreaks vary across regions with different levels of epidemiological risk. For instance, factors influencing outbreaks in Low Risk clusters may differ significantly from those in High Risk clusters, highlighting the need for tailored insights in epidemic modeling.

This research uses COVID-19 pandemic data combined with the climatic and demographic characteristics of various countries and regions as a case study. The goal is to identify different communities with varying levels of epidemiological risk and analyze how climatic and demographic factors influenced COVID-19 outbreaks in each cluster. The findings of this study can help governments, policymakers, and healthcare professionals implement targeted strategies tailored to the specific needs of different communities, enhancing their ability to effectively combat the epidemic. Although this study is driven by the COVID-19 pandemic, its implications go beyond it. The methods and insights presented here can support the broader goal of effectively managing and controlling future pandemics caused by any infectious pathogen.

Our main contributions are as follows:

- We identify distinct communities by applying the fuzzy *c*-means soft clustering technique to a comprehensive dataset that includes daily COVID-19 cases, daily deaths, as well as daily climatic and demographic information from various countries and regions.
- We assess the epidemiological risk associated with each community, providing a clearer understanding of how the impact of the pandemic varies across clusters.
- We extract cluster-specific association rules between climatic and demographic factors and COVID-19 outbreaks in a fuzzy context. These rules provide valuable information, enabling governments and policymakers to implement more targeted and effective preventive measures for different communities.

The remainder of the article is organized as follows. Section 2 presents the state-of-the-art literature. Section 3 outlines the proposed methodology, while Section 4 discusses the experimental results. Finally, Section 5 concludes the article and highlights potential future research directions.

2 Related Work

The growing influence of data science has significantly enhanced the field of epidemiology, enabling more accurate prediction of epidemic outbreaks. Reliable outbreak prediction is essential for helping governments implement effective intervention strategies and allocate public health resources efficiently. At the same time, identifying vulnerable communities and determining key climatic, demographic and economic factors driving pandemic outbreaks provide valuable insights for healthcare professionals. For example, in [7], researchers applied Spectral Clustering (SC), an unsupervised clustering technique, to group countries based on their COVID-19 spread patterns. Similarly, in [6], the authors used Principal Component Analysis (PCA) to reduce the dimensionality of the COVID-19 dataset before applying the *k*-means clustering algorithm. Their results showed that combining PCA with *k*-means produced more accurate community detection outcomes compared to using *k*-means alone. In another study [34], the authors constructed networks in which the nodes represented countries and Canadian provinces, with edges based on travel connections. By applying social network analysis methods along with four different community detection algorithms, they revealed global COVID-19 community structures and examined how these patterns influenced the spread of the virus in Canada.

Despite extensive research in epidemic modeling that highlights the impact of climatic, demographic and economic factors on pandemic outbreaks [3], none of the previously mentioned studies incorporated these factors into their analysis. In [21], the authors addressed this limitation by first applying the Walktrap algorithm to identify clusters of Italian regions with similar COVID-19 patterns. They then integrated climatic data into the analysis, uncovering new community structures. However, their study was limited to a single country and did not involve multi-country analysis together. Moreover, soft clustering has recently gained popularity due to its ability to capture patterns in overlapping data, making it more effective than traditional hard clustering methods [29]. In this context, Mahmoudi et al. [20] conducted an interesting study in which they applied fuzzy clustering to compare the spread of

COVID-19 across the United States, Spain, Italy, Germany, the United Kingdom, France and Iran. In another study [19], researchers used the fuzzy *c*-means clustering method to analyze the epidemiological situations of various European countries, effectively identifying changes in their pandemic trends.

Building on this previous research, our study takes a more comprehensive approach. We not only apply the fuzzy *c*-means clustering algorithm to group countries and regions based on COVID-19 statistics, but also enhance the analysis by incorporating various climatic and demographic factors into the dataset. To gain deeper insights, we further apply fuzzy association rule mining to uncover key climatic and demographic factors driving COVID-19 outbreaks in each cluster. Although some studies [31] in epidemiology have used association rule mining to predict disease dynamics, they have not applied it in a fuzzy framework. For example, a study [14] used association rule mining to analyze the effects of lockdowns, face mask mandates, and mobility patterns on COVID-19 infection rates across 50 states in the United States. However, Buczak et al. [1] adopted fuzzy association rule mining to identify key environmental and climatic factors that influence malaria outbreaks in South Korea. In another work [5], the authors explored the relationships between the climatic and sociodemographic characteristics that affect the spread of COVID-19 using fuzzy association rule mining. However, in this current study, we expand the scope by including a larger number of countries and regions and incorporating a broader set of climatic factors, such as average temperature, relative humidity, precipitation, cloud cover, dew point, and wind speed on a daily basis. Furthermore, after identifying epidemiological risk across different clusters of countries and regions, we extract cluster-specific fuzzy association rules between climatic and demographic factors and COVID-19 outbreaks. This combined approach offers deeper insights into the drivers of pandemic spread across different communities, providing valuable information for policymakers and healthcare professionals to implement more targeted intervention strategies.

3 Proposed Methodology

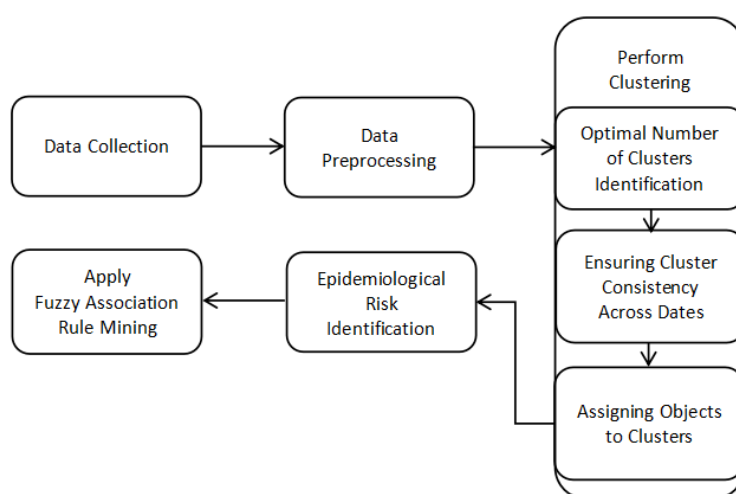


Fig. 1: Overview of the proposed methodology.

Our methodological approach is illustrated in Figure 1, with a detailed explanation of each step provided below.

3.1 Data Collection

The dataset includes daily COVID-19 cases and deaths from April 1, 2020, to June 30, 2020, along with weather and climatic data such as average temperature, relative humidity, precipitation, cloud cover, dew point, and wind speed for 224 countries and regions. Data were sourced from (<https://www.kaggle.com/datasets/eeemonts/weatherclimate-data-covid19/data>). Additionally, population density for each country/region was incorporated.

3.2 Data Preprocessing

Each country or region is represented by nine attributes, including COVID-19 statistics, climatic factors, and demographic information. Due to the varying ranges of these attributes, normalization is crucial to prevent any single feature from dominating the clustering process and to ensure balanced contributions from all attributes. To achieve this, we apply Z-score normalization [26], which standardizes all features to a common scale.

3.3 Perform Clustering

Clustering is a technique used to group objects into clusters such that objects within the same cluster are more similar to each other compared to objects in different clusters. The goal is to maximize the similarity within clusters (intra-cluster similarity) and maximize the dissimilarity between clusters (inter-cluster dissimilarity). A commonly used clustering method is k -means, which assigns each of the n objects to one of the k disjoint clusters based on their proximity to the nearest cluster center, measured by a specific distance metric. However, epidemiological data often show patterns that overlap due to the interconnected influence of climatic and demographic factors. In such cases, fuzzy clustering is more effective as it assigns partial membership of each object to multiple clusters, capturing the uncertainty and overlap in the data. Therefore, in this work, we apply fuzzy c -means (FCM), the most widely used fuzzy clustering algorithm, separately for each date to group countries or regions into k clusters with varying degrees of membership.

Let there be n objects (countries or regions), each characterized by p features. We define the set of clustering objects as $X = \{x_1, x_2, \dots, x_n\}$ with dimensionality $n \times p$ and the set of k cluster centroids as $V = \{v_1, v_2, \dots, v_k\}$ with dimensionality $k \times p$. The membership matrix is denoted by $U = [u_{ij}]_{n \times k}$ where $u_{ij} \in [0, 1]$ represents the membership of the object x_i in cluster j with centroid v_j .

The objective of FCM is to minimize the following function [36]:

$$J(U, V, X) = \sum_{i=1}^n \sum_{j=1}^k u_{ij}^m \|x_i - v_j\|^2 \quad (1)$$

Subject to:

$$\sum_{j=1}^k u_{ij} = 1, \quad \forall i = 1, 2, \dots, n, \quad (2)$$

$$0 < \sum_{i=1}^n u_{ij} < n, \quad \forall j = 1, 2, \dots, k. \quad (3)$$

Here, $m > 1$ is the fuzzifier parameter that controls the degree of fuzziness in the clustering, with $m = 2$ used in this case. The term $\|x_i - v_j\|$ denotes the Euclidean distance between the object x_i and the cluster centroid v_j .

The FCM algorithm iteratively updates the membership values and cluster centroids using the following equations:

$$u_{ij} = \frac{1}{\sum_{c=1}^k \left(\frac{\|x_i - v_j\|}{\|x_i - v_c\|} \right)^{\frac{2}{m-1}}} \quad (4)$$

$$v_j = \frac{\sum_{i=1}^n u_{ij}^m \cdot x_i}{\sum_{i=1}^n u_{ij}^m} \quad (5)$$

The process continues until the centroids converge to stable values or the difference between consecutive iterations becomes smaller than a predefined threshold.

To apply FCM for clustering countries or regions based on COVID-19 statistics, climatic and demographic factors on multiple dates individually, it is essential to determine the optimal number of clusters to ensure the uniformity and comparability of the clustering results.

3.3.1 Optimal Number of Clusters Identification To determine the optimal number of clusters, we use the Xie-Beni Index (XB), a widely recognized cluster validity measure introduced by Xie and Beni in 1991. This index evaluates the quality of clustering considering both the compactness of the clusters and the separation between them.

For the fuzzy *c*-means (FCM) algorithm, the Xie-Beni index is defined as [36]:

$$XB = \frac{J(U, V, X)}{n \times \min_{i \neq j} \|v_i - v_j\|^2} \quad (6)$$

A lower XB value indicates better clustering performance, as it reflects more compact and well-separated clusters. Since clustering is performed separately for each date, our goal is to identify a single optimal number of clusters k that ensures the best cluster quality across all dates. The step-by-step approach is as follows:

1. For k in the range of 2 to 10:

- (a) Set total_xb = 0
- (b) **For each date:**
 - Apply fuzzy *c*-means (FCM) clustering to the feature set of objects (countries or regions) with k clusters.
 - Calculate the Xie-Beni (XB) index using Equation 6.
 - Add the XB index to total_xb.
- (c) **Average the total XB score across all dates:**

$$\text{average_xb} = \frac{\text{total_xb}}{\text{number of dates}}$$

2. Identify the optimal k value that yields the minimum average_xb.

3.3.2 Ensuring Cluster Consistency Across Dates When applying the fuzzy *c*-means (FCM) clustering algorithm to data for different dates, the clusters assigned by FCM are independent for each date. This independence arises because FCM initializes cluster centers randomly and performs optimization separately for each date. As a result, the cluster labels for each date have no correspondence with the labels for the previous date. Due to arbitrarily assigned cluster labels, label inconsistency arises across dates, making it difficult to consistently identify High Risk or Low Risk groups of countries or regions from an epidemiological perspective. A basic solution to address this issue is to use first-date clusters as a reference. For each subsequent date, the Euclidean distances between the current cluster centers and the reference cluster centers are calculated. Each current cluster is then assigned to the closest reference cluster by selecting the center of the reference cluster with the minimum distance. However, this approach has limitations. If two current cluster centers are both closest to the same reference cluster center, the mapping assigns both current clusters to the same reference cluster, leading to duplicate assignments. Furthermore, this method only considers local minimum distances, which can result in suboptimal mappings when distances overlap. To optimally solve the cluster alignment problem, the Hungarian algorithm [15] (also known as the Munkres-Kuhn algorithm) is applied. Introduced by Harold W. Kuhn in 1955, the Hungarian algorithm is an optimization algorithm designed to solve the assignment problem. It efficiently assigns n workers to n tasks (or, in this case, k current clusters to k reference clusters) while minimizing total cost. This ensures a globally optimal one-to-one mapping, preventing duplicate assignments and ensuring consistent cluster labeling across dates. The step-by-step approach to ensure consistent cluster labeling using the Hungarian algorithm is as follows:

1. Initialize FCM clusters for the first date → Reference clusters

2. For each subsequent date:

- (a) Perform FCM clustering → Current clusters
- (b) Compute the distance matrix D between current and reference cluster centers:
 - For k clusters, the size of D is $k \times k$, where each entry is given by: $D(i, j) = \|v_i - v_j\|$. Here, v_i is the centroid of the cluster i on the current date and v_j is the centroid of the cluster j from the reference date.
- (c) Apply the Hungarian algorithm:
 - Find the optimal one-to-one mapping that minimizes the total distance.
- (d) Assign consistent labels according to the reference clusters.

3.3.3 Assigning Objects to Clusters Once the cluster labels are consistent, each object is assigned to a cluster based on its highest membership value. In FCM, each object has a degree of membership in multiple clusters, but it is assigned to the cluster with the highest membership value. After successful assignment, each cluster contains a matrix of objects represented by $[x_{id}]$, where x_{id} denotes the i^{th} country or region on a given date d . Here, i is a subset of n , representing the total number of countries or regions, and d is a subset of all dates.

3.4 Epidemiological Risk Identification

To identify the epidemic risk of each cluster, we analyze the matrix $[x_{id}]$, where each object x_{id} is characterized by p attributes. Among these attributes, we focus on two key indicators: daily COVID-19 cases and daily COVID-19 deaths in various countries or regions on different dates. These two attributes are chosen because they effectively reflect the severity and spread of the pandemic, making them crucial indicators for evaluating the epidemic conditions of the clusters. For each cluster, the mean of daily COVID-19 cases and the mean of daily deaths are calculated by averaging the values of all objects belonging to that cluster. By comparing these two mean values among the different clusters, we can effectively distinguish their epidemic states. Clusters with higher averages represent regions with more severe outbreaks, while those with lower averages correspond to less affected areas. This approach allows for a clear classification of clusters into varying levels of severity of the epidemic.

3.5 Fuzzy Association Rule Mining

After identifying the clusters, our objective is to discover the association rules specific to the cluster that reveal how climatic and demographic factors influence COVID-19 outbreaks in each cluster. In this process, the transaction set T contains all objects present in a cluster with a membership value greater than or equal to 0.5. The itemset p includes COVID-19 statistics, climatic and demographic information that characterize each transaction in the transaction set. Since the items of the itemset p are by nature numeric or continuous, the application of a standard association rule mining algorithm is not feasible. Therefore, the first task is to convert the quantitative attributes into categories (for example, Low, Medium, and High). An effective approach is to define linguistic variables using fuzzy sets. After this transformation, we perform fuzzy association rule mining [10], where the rules follow the following format: $(R \text{ is } A_R) \rightarrow (S \text{ is } C_S)$. Here, $R, S \subseteq p$ and the itemset $R = \{r_1, r_2, \dots, r_a\}$ represents the antecedent, while the itemset $S = \{s_1, s_2, \dots, s_b\}$ forms the consequent. The fuzzy categories associated with the items in the antecedent are denoted by $A_R = \{F_{r1}, F_{r2}, \dots, F_{ra}\}$ and those for the consequent are represented by $C_S = \{F_{s1}, F_{s2}, \dots, F_{sb}\}$. These fuzzy categories are defined by membership functions that assign linguistic labels (such as Low, Medium, or High) to the corresponding items in R and S . The steps for extracting important fuzzy association rules for each cluster are as follows:

Fuzzification: In this step, the quantitative values of different attributes for each transaction in T are converted into fuzzy sets using membership functions, as shown in Figure 2. For each attribute, appropriate threshold values are determined by analyzing its data distribution to classify the values into linguistic labels such as Low, Medium, and High.

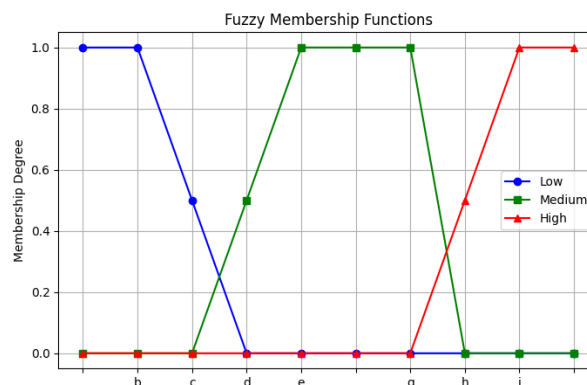


Fig. 2: Fuzzy membership functions with linguistic labels for fuzzification.

Fuzzy Normalization: After fuzzification, the membership values of each attribute are normalized for every transaction. This is done by dividing the membership value of an attribute in one fuzzy class by the total membership values of that attribute across all its fuzzy classes. This step ensures that the membership values of each attribute across its fuzzy classes in a transaction sum to 1.

Support Calculation for 1-Itemsets and Identification of Frequent 1-Itemsets: To calculate the support for 1-itemsets, the total normalized membership value for each fuzzy class is divided by the total number of transactions. If the support of any 1-itemset is greater than or equal to a predefined minimum support value, it is classified as a frequent 1-itemset.

Frequent k -Itemsets Generation: Frequent k -itemsets are generated from frequent $(k - 1)$ -itemsets. To determine the support of each k -itemset:

- Identify the smallest normalized membership value among the k fuzzy classes (which form the k -itemset) in each transaction.
- Sum these minimum values across all transactions.
- Divide the resulting sum by the total number of transactions.
- If the support of a k -itemset meets or exceeds the minimum support threshold, it is considered a frequent k -itemset. Otherwise, it is rejected.

This step is repeated iteratively until no more frequent itemsets can be generated.

Reliable Rule Extraction: From each k -itemset ($k \geq 2$), all possible association rules are extracted. For each rule, the confidence and lift are calculated [16]. The confidence of a rule is expressed as : $\frac{\text{Support}(\text{Antecedent} \cup \text{Consequent})}{\text{Support}(\text{Antecedent})}$. The lift of a rule is defined as : $\frac{\text{Confidence}}{\frac{\text{Support}(\text{Consequent})}{|T|}}$. Association rules are regarded reliable and significant only if their confidence value meets or exceeds the minimum confidence threshold and their lift value is greater than 1.

Identifying Interesting Association Rules: Finally, we retain only those rules where COVID-19 cases and COVID-19 deaths appear in the consequent part. This is because our goal is to identify the climatic and demographic factors that contribute to the COVID-19 outbreak in a specific cluster.

4 Experimental Results

The dataset comprises 224 countries and various regions within certain countries. For each country or region, the dataset records population density, daily COVID-19 cases and deaths, and climatic factors such as daily average temperature, relative humidity, precipitation, cloud cover, dew point, and wind speed, spanning from April 1, 2020, to June 30, 2020.

To analyze the data, a date-wise clustering approach was employed using the fuzzy c -Means (FCM) algorithm. The optimal number of clusters was determined to be four, based on the Xie-Beni index, ensuring a well-balanced clustering structure. Tables 1 and 2 illustrate the temporal evolution of the cluster membership values for India and Afghanistan, respectively, over a specific period of time. The final cluster assignment for each day is determined based on the highest membership value. A key observation from Table 2 is that Afghanistan remained in Cluster_1 with the highest membership value from April 5, 2020, to April 12, 2020. In contrast, Table 1 reveals that India's cluster assignment changed multiple times during the same period. This suggests that the epidemiological situation in Afghanistan was more stable, while India experienced more fluctuations in its risk classification.

Figures 3, 4, and 5 visualize the spatial distribution of clusters across different countries and regions for April 1, April 2, and June 30, 2020. These figures highlight the dynamic nature of clustering, as the countries or regions assigned to each cluster, along with their corresponding membership values, change over time.

Each of the four clusters consists of specific countries or regions, assigned based on the highest degree of membership for each day between April 1 and June 30, 2020. To evaluate the epidemiological risk levels associated with each cluster, we analyze the average daily COVID-19 cases and average daily COVID-19 deaths of the countries or regions within each cluster. Table 3 presents these averages, revealing that Cluster_2 has the lowest average daily cases and deaths, indicating a Very Low Risk level. In contrast, Cluster_3 exhibits the highest values for both metrics, categorizing it as a High Risk cluster. Based on

these observations, we classify the clusters as follows: Cluster_1 as Moderate Risk, Cluster_2 as Very Low Risk, Cluster_3 as High Risk, and Cluster_4 as Low Risk.

The epidemiological risk classification is further validated using the fuzzy association rule mining technique. Specifically, the support values for the attributes cases_Low and deaths_Low, as presented in Table 4, provide additional confirmation of the cluster identification. Cluster_2, classified as Very Low Risk, has the highest support values for cases_Low and deaths_Low. Furthermore, cases_Medium and deaths_Medium do not appear as frequent 1-itemsets, reinforcing its classification as a Very Low Risk cluster. In contrast, Cluster_3 has the lowest support values for cases_Low and deaths_Low, while achieving the highest support values for cases_Medium and deaths_Medium, further validating its High Risk status.

To identify important cluster-specific rules that link various climatic and demographic factors with COVID-19 data, we employ the Fuzzy Association Rule Mining Technique. We set the minimum support threshold at 0.1 and the minimum confidence threshold at 0.3. Only rules that meet the criteria of $confidence \geq 0.3$ and $lift > 1$ are retained. The significant rules extracted from frequent 2-itemsets for Cluster_1, Cluster_2, Cluster_3, and Cluster_4 are presented in Tables 5, 6, 7, and 8 respectively.

This analysis highlights the key climatic and demographic factors influencing epidemic outbreaks in each cluster. For example, cloud cover does not directly contribute to COVID-19 outbreaks in Cluster_2 (Table 6), whereas it has a significant impact on the other three clusters (Tables 5, 7, and 8). Similarly, relative humidity does not directly influence COVID-19 outbreaks in Cluster_4 (Table 8), but it plays a crucial role in the remaining clusters (Tables 5, 6, and 7). However, when cloud cover or relative humidity interacts with other climatic or demographic factors, it influences COVID-19 trends, as evidenced in Tables 10 and 12, respectively. In addition, the association rules provide valuable validation for the classification of cluster-based epidemiological risk. Cluster_2 (Very Low Risk) exhibits the strongest and most consistent associations with low case numbers, supported by high confidence values (0.96–1.00) and the complete absence of rules for medium case levels, as shown in Table 6. In contrast, Cluster_1 (Moderate Risk) demonstrates a relationship between medium cloud cover and medium case levels (Table 5), indicating some variability in case numbers. In particular, Cluster_3 (High Risk) is the only cluster where medium case and death levels frequently appear as the consequent in the association rules, as seen in Tables 7 and 11, further confirming its greater epidemiological risk.

Tables 9, 10, 11, and 12 present a subset of key association rules derived from frequent 3-itemsets for Clusters_1, 2, 3, and 4, respectively, revealing how different combinations of climatic and demographic factors influence COVID-19 trends within each cluster. Similarly, Tables 13, 14, 15, and 16 provide significant rules from frequent 8-itemsets, representing more complex relationships between climatic and demographic attributes and COVID-19 outbreaks. These higher order combinations allow for a deeper understanding of how multiple environmental and demographic factors jointly impact COVID-19 statistics. A notable distinction is that only Cluster_2 (Very Low Risk) has a rule generated from a frequent 9-itemset, as presented in Table 17, suggesting that a larger combination of factors contributes to its epidemiological trends. In contrast, for Clusters_1, 3, and 4, the highest level of rule generation occurs at frequent 8-itemsets, meaning that beyond eight combined attributes, no additional meaningful patterns emerge. Overall, this analysis provides critical insights into how climatic and demographic factors interact differently across clusters, revealing complex dependencies that contribute to varying COVID-19 risk levels.

Table 1: Date-wise cluster membership values and final cluster assignment for India.

Date	Location	Cluster_1	Cluster_2	Cluster_3	Cluster_4	Cluster Assignment
04-01-2020	Country - India	0.09	0.49	0.13	0.29	Cluster_2
04-02-2020	Country - India	0.13	0.41	0.25	0.20	Cluster_2
04-03-2020	Country - India	0.12	0.54	0.19	0.14	Cluster_2
04-04-2020	Country - India	0.11	0.17	0.07	0.65	Cluster_4
04-05-2020	Country - India	0.12	0.66	0.08	0.14	Cluster_2
04-06-2020	Country - India	0.15	0.56	0.16	0.13	Cluster_2
04-07-2020	Country - India	0.22	0.40	0.15	0.23	Cluster_2
04-08-2020	Country - India	0.24	0.39	0.24	0.13	Cluster_2
04-09-2020	Country - India	0.19	0.28	0.39	0.14	Cluster_3
04-10-2020	Country - India	0.15	0.39	0.30	0.16	Cluster_2
04-11-2020	Country - India	0.13	0.31	0.24	0.33	Cluster_4
04-12-2020	Country - India	0.25	0.30	0.31	0.15	Cluster_3

Table 2: Date-wise cluster membership values and final cluster assignment for Afghanistan.

Date	Location	Cluster_1	Cluster_2	Cluster_3	Cluster_4	Cluster Assignment
04-01-2020	Country - Afghanistan	0.43	0.11	0.34	0.11	Cluster_1
04-02-2020	Country - Afghanistan	0.24	0.11	0.56	0.10	Cluster_3
04-03-2020	Country - Afghanistan	0.60	0.08	0.23	0.09	Cluster_1
04-04-2020	Country - Afghanistan	0.24	0.18	0.46	0.12	Cluster_3
04-05-2020	Country - Afghanistan	0.40	0.13	0.27	0.21	Cluster_1
04-06-2020	Country - Afghanistan	0.45	0.13	0.24	0.18	Cluster_1
04-07-2020	Country - Afghanistan	0.41	0.12	0.34	0.13	Cluster_1
04-08-2020	Country - Afghanistan	0.44	0.14	0.26	0.16	Cluster_1
04-09-2020	Country - Afghanistan	0.45	0.15	0.19	0.20	Cluster_1
04-10-2020	Country - Afghanistan	0.44	0.15	0.21	0.21	Cluster_1
04-11-2020	Country - Afghanistan	0.39	0.25	0.21	0.16	Cluster_1
04-12-2020	Country - Afghanistan	0.32	0.17	0.22	0.29	Cluster_1

Table 3: Epidemiological risk classification for different clusters.

Cluster	Average daily COVID-19 cases	Average daily COVID-19 deaths	Risk Level
Cluster_1	519.40	32.64	Moderate Risk
Cluster_2	152.70	5.26	Very Low Risk
Cluster_3	805.37	38.94	High Risk
Cluster_4	268.82	12.48	Low Risk

Table 4: Frequent 1-itemsets with their support values for Cluster_1, Cluster_2, Cluster_3, and Cluster_4.

Cluster_1		Cluster_2		Cluster_3		Cluster_4	
Itemset	Support	Itemset	Support	Itemset	Support	Itemset	Support
cases_Low	0.72	cases_Low	0.96	cases_Low	0.62	cases_Low	0.80
cases_Medium	0.27	deaths_Low	1.00	cases_Medium	0.34	cases_Medium	0.20
deaths_Low	0.93	temp_Medium	0.87	deaths_Low	0.90	deaths_Low	0.96
temp_Low	0.54	temp_High	0.12	deaths_Medium	0.10	temp_Low	0.10
temp_Medium	0.45	humi_Medium	0.13	temp_Low	0.34	temp_Medium	0.83
humi_Medium	0.30	humi_High	0.87	temp_Medium	0.46	humi_High	0.94
humi_High	0.70	precip_Low	0.79	temp_High	0.20	precip_Low	0.14
precip_Low	0.33	precip_Medium	0.21	humi_Low	0.39	precip_Medium	0.35
precip_Medium	0.27	PD_Low	0.25	humi_Medium	0.60	precip_High	0.51
precip_High	0.41	PD_Medium	0.57	precip_Low	0.99	PD_Low	0.45
PD_Low	0.47	PD_High	0.19	PD_Low	0.50	PD_Medium	0.48
PD_Medium	0.47	cloud_Low	0.36	PD_Medium	0.43	cloud_Medium	0.35
cloud_Medium	0.36	cloud_Medium	0.59	cloud_Low	0.74	cloud_High	0.62
cloud_High	0.59	dew_Medium	0.13	cloud_Medium	0.25	dew_Medium	0.30
dew_Low	0.15	dew_High	0.87	dew_Low	0.49	dew_High	0.70
dew_Medium	0.56	wind_Low	0.22	dew_Medium	0.51	wind_Low	0.89
dew_High	0.29	wind_Medium	0.65	wind_Low	0.67	wind_Medium	0.11
wind_Low	0.85	wind_High	0.13	wind_Medium	0.33		
wind_Medium	0.15						

Table 9: Some interesting rules generated from frequent 3-itemsets of Cluster_1.

Rule	Antecedent	Consequent	Confidence	Lift
1	{dew_High}	{cases_Low, deaths_Low}	0.72	1.01
2	{cloud_High, precip_High}	{deaths_Low}	0.94	1.01
3	{PD_Medium, cloud_High}	{cases_Low}	0.74	1.03
4	{humi_High, precip_Medium}	{deaths_Low}	0.94	1.01
5	{PD_Low, precip_High}	{deaths_Low}	0.94	1.01
6	{PD_Low, humi_Medium}	{deaths_Low}	0.94	1.01
7	{cloud_High, temp_Low}	{cases_Low}	0.74	1.03
8	{precip_Medium, temp_Medium}	{deaths_Low}	0.96	1.03
9	{precip_Medium, wind_Low}	{deaths_Low}	0.94	1.01
10	{humi_High, wind_Low}	{cases_Low}	0.73	1.02
11	{cloud_High, precip_Medium}	{cases_Low}	0.74	1.03
12	{dew_High, precip_High}	{deaths_Low}	0.95	1.02
13	{PD_Medium, humi_High}	{cases_Low}	0.73	1.02
14	{PD_Low, wind_Low}	{deaths_Low}	0.94	1.01
15	{dew_High, precip_High}	{cases_Low}	0.73	1.02

Table 5: Interesting rules generated from frequent 2-itemsets of Cluster_1.

Rule	Antecedent	Consequent	Confidence	Lift
1	{humi_High}	{deaths_Low}	0.94	1.01
2	{humi_High}	{cases_Low}	0.73	1.02
3	{PD_Low}	{deaths_Low}	0.94	1.01
4	{cloud_High}	{cases_Low}	0.74	1.03
5	{cloud_Medium}	{cases_Medium}	0.31	1.13
6	{cloud_High}	{deaths_Low}	0.94	1.01
7	{temp_Medium}	{deaths_Low}	0.96	1.03
8	{wind_Low}	{deaths_Low}	0.94	1.01
9	{wind_Low}	{cases_Low}	0.73	1.02
10	{dew_High}	{deaths_Low}	0.95	1.03
11	{dew_High}	{cases_Low}	0.72	1.01
12	{dew_Medium}	{cases_Low}	0.73	1.02
13	{precip_Medium}	{deaths_Low}	0.94	1.01
14	{precip_High}	{deaths_Low}	0.94	1.01
15	{precip_Medium}	{cases_Low}	0.72	1.01
16	{precip_High}	{cases_Low}	0.73	1.01

Table 6: Interesting rules generated from frequent 2-itemsets of Cluster_2.

Rule	Antecedent	Consequent	Confidence	Lift
1	{wind_Medium}	{cases_Low}	0.97	1.02
2	{PD_High}	{cases_Low}	0.99	1.03
3	{wind_High}	{cases_Low}	1.00	1.05
4	{temp_Medium}	{cases_Low}	0.96	1.01
5	{humi_High}	{cases_Low}	0.97	1.01
6	{dew_High}	{cases_Low}	0.97	1.01

Table 7: Interesting rules generated from frequent 2-itemsets of Cluster_3.

Rule	Antecedent	Consequent	Confidence	Lift
1	{cloud_Medium}	{cases_Low}	0.66	1.07
2	{cloud_Low}	{cases_Medium}	0.35	1.03
3	{cloud_Medium}	{deaths_Low}	0.93	1.04
4	{PD_Low}	{deaths_Low}	0.91	1.01
5	{PD_Low}	{cases_Low}	0.63	1.02
6	{PD_Medium}	{cases_Medium}	0.40	1.19
7	{dew_Medium}	{deaths_Low}	0.91	1.02
8	{dew_Medium}	{cases_Low}	0.64	1.03
9	{dew_Medium}	{cases_Medium}	0.34	1.01
10	{wind_Medium}	{cases_Medium}	0.38	1.12
11	{wind_Low}	{cases_Low}	0.64	1.03
12	{temp_Low}	{deaths_Low}	0.92	1.03
13	{temp_Low}	{cases_Low}	0.64	1.03
14	{temp_Medium}	{cases_Low}	0.67	1.08
15	{temp_Medium}	{deaths_Low}	0.90	1.01
16	{temp_Low}	{cases_Medium}	0.35	1.04
17	{humi_Low}	{cases_Medium}	0.35	1.04
18	{humi_Medium}	{cases_Low}	0.66	1.06
19	{humi_Medium}	{deaths_Low}	0.93	1.03

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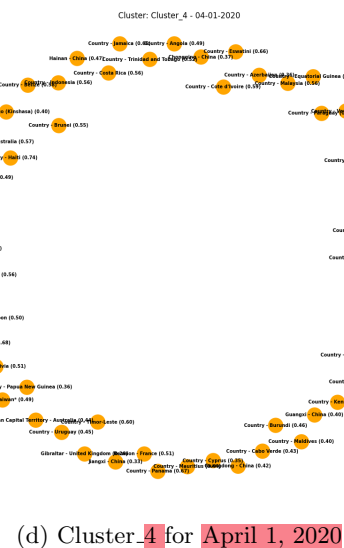
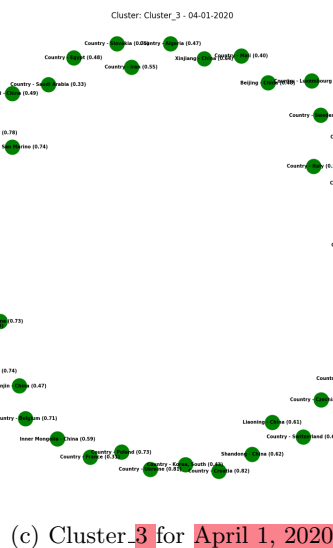
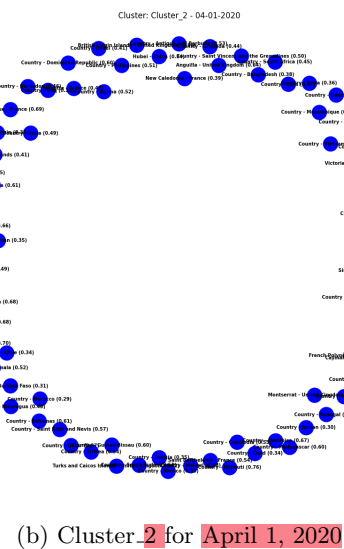
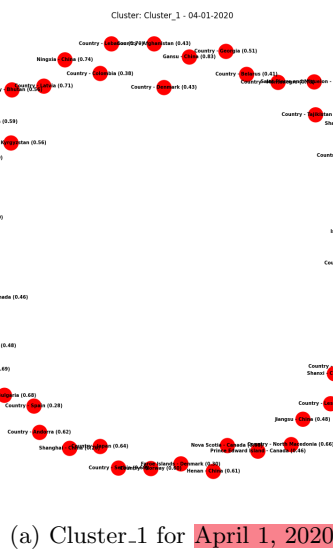


Fig. 3: Clustering results using FCM for April 1, 2020, displayed in four subfigures: (a), (b), (c), and (d).

5 Conclusion

In this study, we analyzed COVID-19 data together with various climatic and demographic information from numerous countries and regions throughout the world, grouping them into clusters daily from April 1 to June 30, 2020. We applied the fuzzy *c*-means clustering technique, which allows regions to belong to multiple clusters simultaneously, with one cluster generally being dominant. This approach provides a more flexible and realistic representation of epidemiological patterns. Beyond clustering, we assessed the epidemiological risk levels of each group and identified key environmental and demographic factors influencing virus transmission within each cluster using fuzzy association rule mining. Our findings reveal different cluster-specific relationships that can help governments, policymakers, and healthcare professionals develop targeted strategies to manage the pandemic more effectively.

For future research, our aim is to improve the reliability of association rules by incorporating a time window-based model. This approach will identify consistent rules that hold across different time periods, providing a more stable understanding of the links between climatic, demographic factors, and disease outbreaks. Additionally, integrating time-series forecasting models could help predict future association rules, offering valuable insights to support informed decision making and improve preparedness for future pandemics.

Table 8: Interesting rules generated from frequent 2-itemsets of Cluster_4.

Rule	Antecedent	Consequent	Confidence	Lift
1	{PD_Medium}	{cases_Low}	0.81	1.02
2	{PD_Medium}	{deaths_Low}	0.97	1.01
3	{dew_High}	{deaths_Low}	0.97	1.01
4	{dew_High}	{cases_Low}	0.80	1.01
5	{precip_Low}	{deaths_Low}	0.97	1.01
6	{precip_Low}	{cases_Low}	0.82	1.03
7	{precip_Medium}	{cases_Low}	0.82	1.02
8	{precip_Medium}	{deaths_Low}	0.97	1.01
9	{wind_Medium}	{deaths_Low}	0.99	1.02
10	{temp_Medium}	{cases_Low}	0.80	1.01
11	{cloud_Medium}	{deaths_Low}	0.97	1.01
12	{cloud_Medium}	{cases_Low}	0.80	1.01

Table 10: Some interesting rules generated from frequent 3-itemsets of Cluster_2.

Rule	Antecedent	Consequent	Confidence	Lift
1	{cloud_Low, wind_Medium}	{cases_Low}	0.97	1.01
2	{humi_High}	{cases_Low, deaths_Low}	0.97	1.01
3	{humi_High, temp_Medium}	{cases_Low}	0.97	1.02
4	{PD_High, temp_Medium}	{cases_Low}	0.99	1.04
5	{dew_High}	{cases_Low, deaths_Low}	0.97	1.01
6	{wind_Medium}	{cases_Low, deaths_Low}	0.97	1.02
7	{cloud_Low, temp_Medium}	{cases_Low}	0.96	1.01
8	{PD_Medium, humi_High}	{cases_Low}	0.96	1.01
9	{PD_High, humi_High}	{cases_Low}	0.99	1.04
10	{PD_Medium, dew_High}	{cases_Low}	0.97	1.01
11	{PD_High, dew_High}	{cases_Low}	0.99	1.03
12	{temp_Medium}	{cases_Low, deaths_Low}	0.96	1.01
13	{PD_Medium, wind_Medium}	{cases_Low}	0.97	1.01
14	{PD_High, wind_Medium}	{cases_Low}	0.99	1.03
15	{dew_High, wind_High}	{cases_Low}	1.00	1.05

Table 11: Some interesting rules generated from frequent 3-itemsets of Cluster_3.

Rule	Antecedent	Consequent	Confidence	Lift
1	{cloud_Low, precip_Low}	{cases_Medium}	0.35	1.03
2	{dew_Medium, precip_Low}	{cases_Medium}	0.34	1.01
3	{cloud_Low, humi_Low}	{cases_Medium}	0.37	1.10
4	{precip_Low, temp_Low}	{cases_Medium}	0.35	1.04
5	{PD_Medium}	{cases_Medium, deaths_Low}	0.32	1.17
6	{PD_Medium, humi_Medium}	{cases_Medium}	0.42	1.22
7	{PD_Medium, cloud_Low}	{cases_Medium}	0.42	1.22
8	{PD_Medium, wind_Low}	{cases_Medium}	0.41	1.20
9	{precip_Low, wind_Medium}	{cases_Medium}	0.38	1.12
10	{PD_Medium, precip_Low}	{cases_Medium}	0.40	1.19
11	{cloud_Low, temp_Low}	{deaths_Low}	0.92	1.03
12	{PD_Medium, temp_Medium}	{cases_Low}	0.63	1.01
13	{humi_Medium, wind_Medium}	{deaths_Low}	0.94	1.04
14	{temp_Medium, wind_Low}	{deaths_Low}	0.91	1.01
15	{precip_Low, temp_Medium}	{cases_Low}	0.67	1.08

Table 12: Some interesting rules generated from frequent 3-itemsets of Cluster_4.

Rule	Antecedent	Consequent	Confidence	Lift
1	{PD_Medium, dew_High}	{deaths_Low}	0.98	1.02
2	{PD_Medium, humi_High}	{cases_Low}	0.82	1.03
3	{cloud_Medium}	{cases_Low, deaths_Low}	0.80	1.01
4	{precip_Medium, temp_Medium}	{cases_Low}	0.81	1.02
5	{humi_High, precip_Medium}	{deaths_Low}	0.97	1.01
6	{cloud_Medium, precip_Medium}	{deaths_Low}	0.98	1.02
7	{cloud_Medium, dew_High}	{cases_Low}	0.81	1.02
8	{PD_Medium, precip_Medium}	{deaths_Low}	0.99	1.03
9	{PD_Medium}	{cases_Low, deaths_Low}	0.81	1.02
10	{cloud_High, precip_Medium}	{cases_Low}	0.81	1.02
11	{PD_Medium, wind_Low}	{cases_Low}	0.80	1.01
12	{dew_High}	{cases_Low, deaths_Low}	0.80	1.01
13	{dew_High, humi_High}	{deaths_Low}	0.97	1.01
14	{cloud_Medium, temp_Medium}	{deaths_Low}	0.97	1.01
15	{PD_Medium, dew_High}	{cases_Low}	0.81	1.02

Table 13: Interesting rule generated from frequent 8-itemset of Cluster_1.

Rule	Antecedent	Consequent	Confidence	Lift
1	{cloud_High, dew_High, humi_High, precip_High, temp_Medium, wind_Low}	{cases_Low, deaths_Low}	0.73	1.02

Table 14: Interesting rules generated from frequent 8-itemsets of Cluster_2.

Rule	Antecedent	Consequent	Confidence	Lift
1	{cloud_Medium, dew_High, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.04
2	{PD_Medium, cloud_Medium, dew_High, humi_High, precip_Low, temp_Medium}	{cases_Low, deaths_Low}	0.98	1.03
3	{PD_Medium, cloud_Low, dew_High, humi_High, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.98	1.03
4	{PD_Medium, cloud_Low, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03
5	{PD_Medium, cloud_Medium, dew_High, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low}	0.99	1.03
6	{cloud_Low, dew_High, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03
7	{PD_Medium, cloud_Medium, dew_High, humi_High, precip_Low, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03
8	{PD_Medium, cloud_Low, dew_High, humi_High, precip_Low, temp_Medium}	{cases_Low, deaths_Low}	0.98	1.03
9	{PD_Medium, dew_High, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03
10	{PD_Medium, cloud_Medium, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.98	1.03
11	{PD_Medium, cloud_Low, dew_High, humi_High, precip_Low, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03
12	{PD_Medium, cloud_Medium, dew_High, humi_High, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.98	1.03
13	{PD_Medium, cloud_Medium, dew_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03

Table 15: Interesting rule generated from frequent 8-itemset of Cluster_3.

Rule	Antecedent	Consequent	Confidence	Lift
1	{cloud_Low, dew_Medium, humi_Medium, precip_Low, temp_Medium, wind_Low}	{cases_Low, deaths_Low}	0.67	1.09

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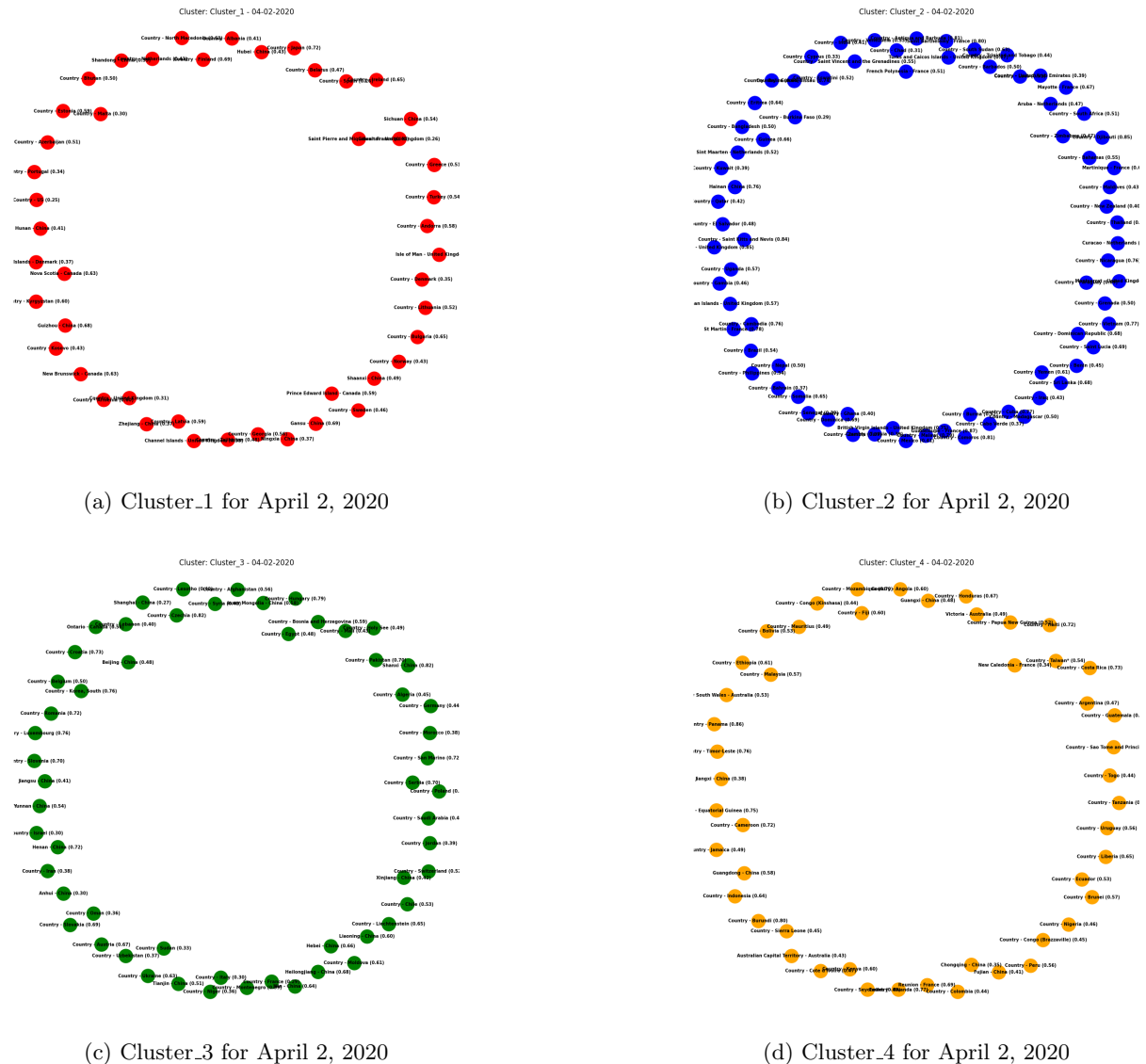


Fig. 4: Clustering results using FCM for April 2, 2020, displayed in four subfigures: (a), (b), (c), and (d).

Table 16: Interesting rules generated from frequent 8-itemsets of Cluster_4.

Rule	Antecedent	Consequent	Confidence	Lift
1	{PD_Medium, cloud_High, dew_High, humi_High, precip_High, temp_Medium}	{cases_Low, deaths_Low}	0.80	1.01
2	{PD_Medium, cloud_High, dew_High, humi_High, precip_High, wind_Low}	{cases_Low, deaths_Low}	0.80	1.01
3	{PD_Medium, cloud_High, humi_High, precip_High, temp_Medium, wind_Low}	{cases_Low, deaths_Low}	0.82	1.03

Table 17: Interesting rule generated from frequent 9-itemset of Cluster_2.

Rule	Antecedent	Consequent	Confidence	Lift
1	{PD_Medium, cloud_Medium, dew_High, humi_High, precip_Low, temp_Medium, wind_Medium}	{cases_Low, deaths_Low}	0.99	1.03

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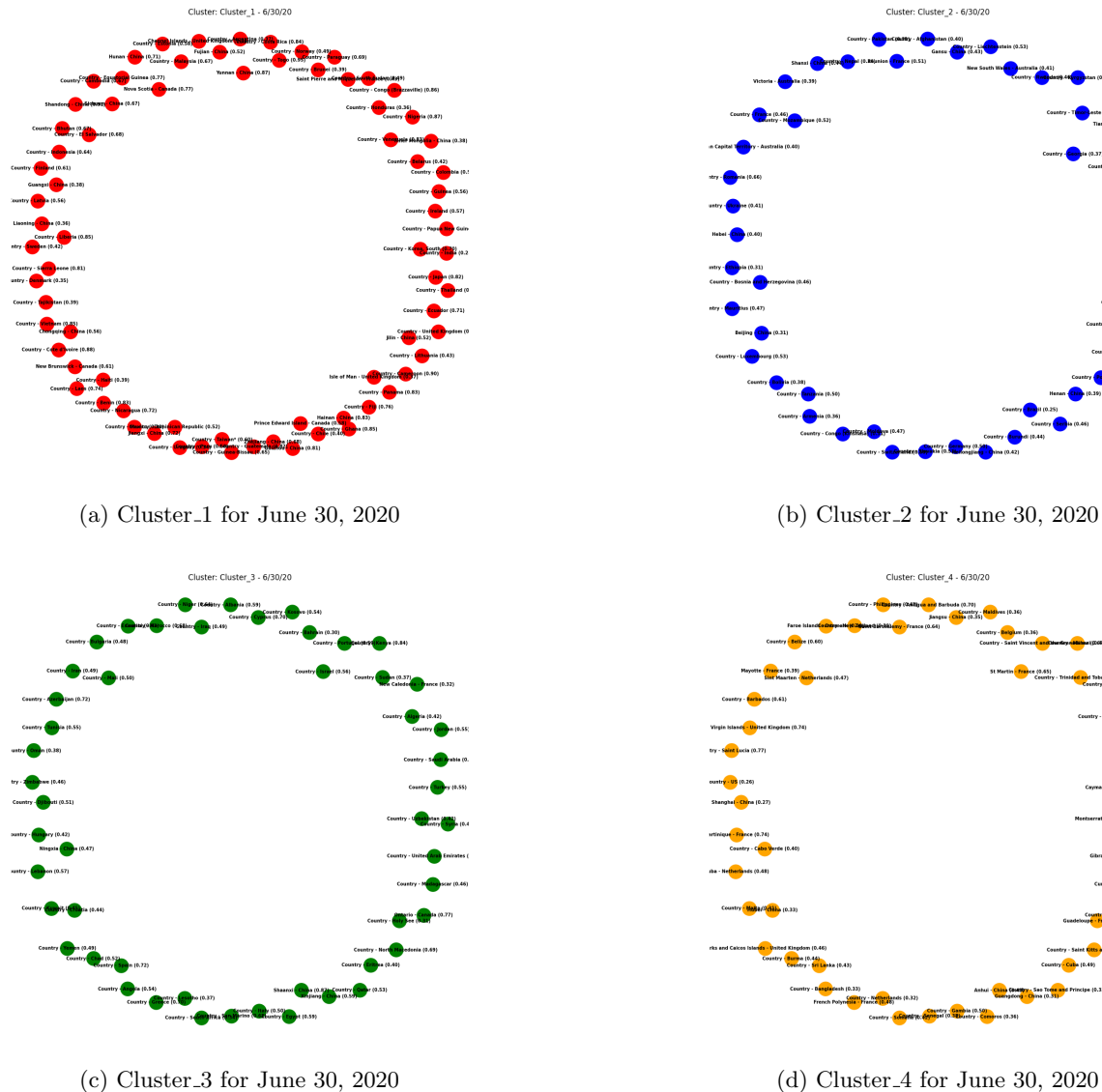


Fig. 5: Clustering results using FCM for June 30, 2020, displayed in four subfigures: (a), (b), (c), and (d).

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